Word Embeddings (II)

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Applications of Natural Language Processing

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With slides by Brendan O'Connor

Word embeddings recap

- Idea: Words with similar meanings appear in similar contexts
- Vector representations of words learned from corpus data
- Vector similarity reflects similarity of observed contexts

Word embeddings recap

- Space of word embeddings forms a meaning space
- Proximity in space indicates similarity in meaning
- Measures:
  - Distance measures, like Euclidean distance: lower value = higher similarity
  - Similarity measures, like Cosine similar: higher value = higher similarity

Euclidean Distance

$$d(x, y) = \sqrt{\sum_i (x_i - y_i)^2}$$

Issue: Vector length depends on frequency. More frequent words will have longer vectors.

Cosine Similarity

$$s(x, y) = \frac{x \cdot y}{|x| \; |y|}$$

Only depends on vector angle

Range: -1 to 1

Nearest neighbors

- Nearest neighbors of word w: among the whole vocabulary, words v with highest cosine similarity sim(w, v)
- Often used as a way of interpreting word embedding

What do word embedding encode?

- What kinds of words come out as similar, and why?

What do word embedding encode?

- What kinds of words come out as similar, and why?
  - Synonyms and near-synonyms:
    amiable / agreeable / congenial
  - “Sister terms”:
    cat / dog
  - Topically related words:
    fire / attack / smoke

What do word embedding encode?

- What do you think happens with antonyms?
  - hot / cold
  - rise / fall

- Words that have more than one meaning: What will they be similar to?
  - orange: fruit or color
  - bat: stick or animal

A neat result: We can distinguish semantic relations in word embeddings

Same corpus (BNC), different window sizes:
Nearest neighbors of "dog"

2-word window:

- cat
- horse
- fox
- pet
- rabbit
- pig

- animal
- mongrel
- sheep
- pigeon

30-word window:

- kennel
- puppy
- pet
- bitch
- terrier
- rottweiler

- canine
- cat
- to bark
- Alsatian

Results by: Marco Baroni

A neat result: We can distinguish semantic relations in word embeddings

Taxonomic similarity, "similarity":
Words that are close together in an ontology,
like "dog" and "cat".;
Especially high cosines for co-hyponyms
Small context windows

Topical similarity, "relatedness":
Words that tend to co-appear in the same stories
Large context windows

Results by: Marco Baroni

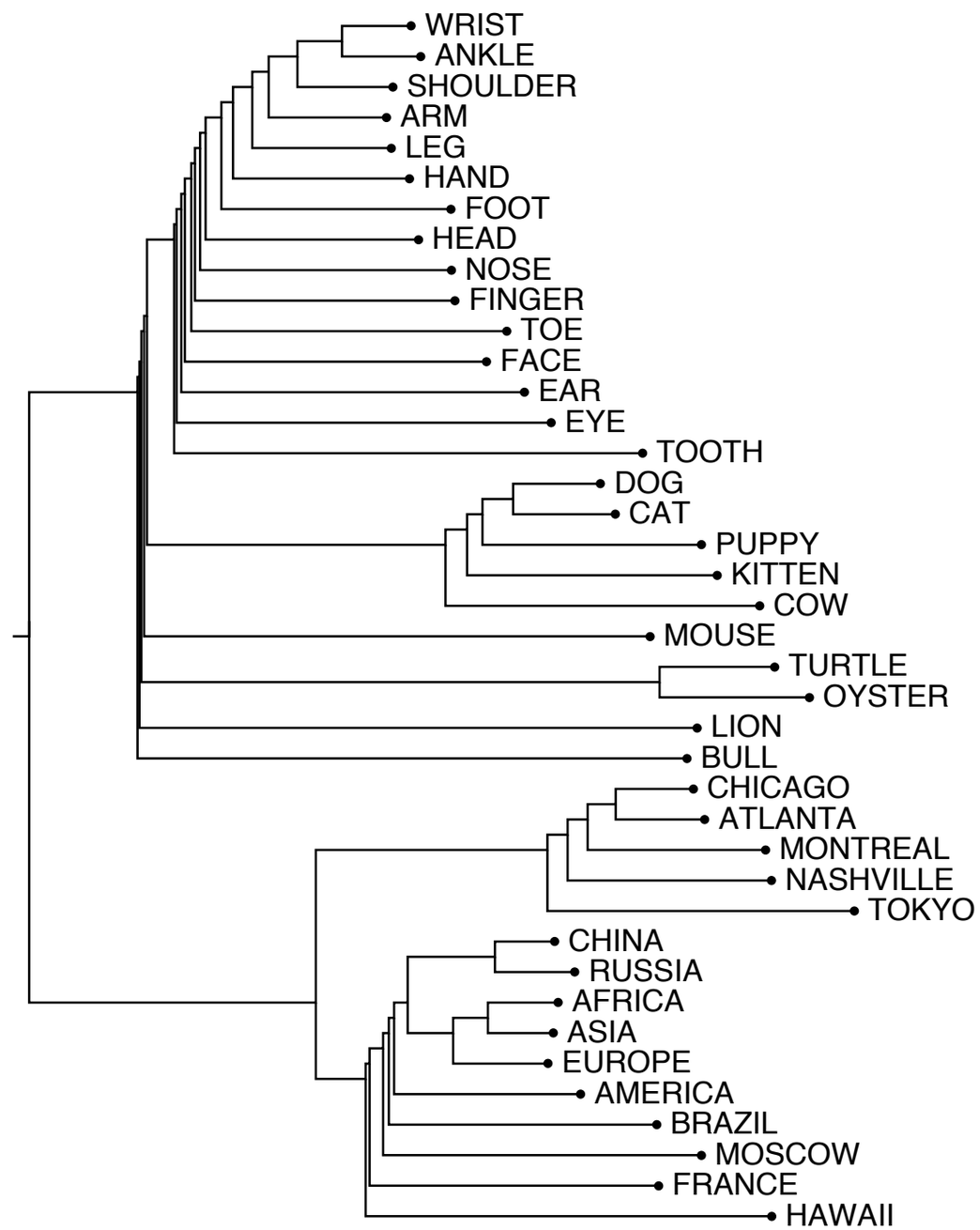
Visualizing embeddings: Downprojecting to 2 dimensions

- Embeddings usually have many dimensions, from 200 to in the 1000s
- How can we visualize the neighborhood of a word?
- Downprojecting to 2 dimensions in a way that preserves proximity

Visualizing embeddings: Downprojecting to 2 dimensions

- Principal Component Analysis (PCA):
  Given a set X of points (embeddings),
  find new set of axes to represent X such that
  - first axis: the direction of greatest variance in the data
  - second axis: direction of greatest variance in the data once first axis has been removed, and so on
- Axes are orthogonal to each other
- Axes are ordered: First axis covers most of the variance in the data
- For visualization, we drop all but the first two axes

Another way to visualize: hierarchical clustering of embeddings



SLP3 ch. 6

What does it learn?

- Demo: GLOVE embedding similarities
  - fasttext, glove, and word2vec are most-often used pretrained word embeddings

Evaluating word embeddings

- How large should your corpus be for you to obtain reasonable word embeddings?
- No clear cut-off
- Some standard evaluations for a sanity check of word embeddings:
  - Matching human ratings of pairwise word similarities (this is the most standard test)
  - Predicting analogies

Approximating human judgments about word similarity

- Small datasets: Rubenstein/Goodenough, WordSim353
- Larger datasets: SimLex, SimVerb, MEN

RG examples

gem	jewel	3.94
automobile	car	3.920
bird	crane	2.630
oracle	sage	2.610
hill	woodland	1.480
crane	rooster	1.410
lad	wizard	0.990
rooster	voyage	0.040
cord	smile	0.020

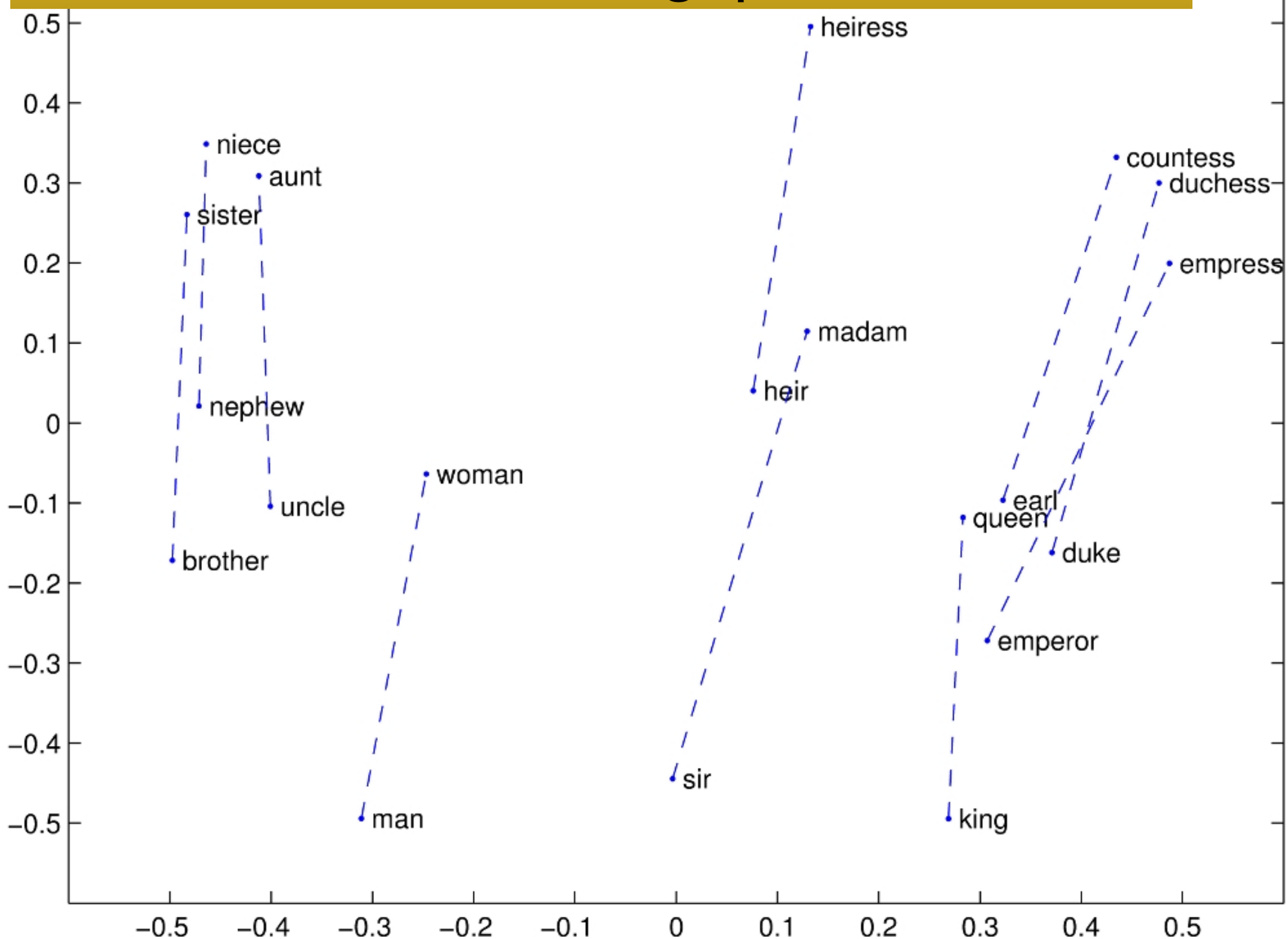
Approximating human judgments about word similarity

- For each word pair from the dataset, compute **cosine** similarity
- Measure **correlation** between human judgments and cosine similarities: Are cosines larger for word pairs with higher human judgments?
- Measuring correlation: Spearman's rho for rank similarity
  Don't assume human ratings are interval data

Approximating human judgments about word similarity

- Separate human datasets for similarity (taxonomic similarity) and relatedness (topical similarity)
  - Taxonomic: SimLex, SimVerb
  - Topical (and visual): MEN

Analogies are (more or less) constant offsets in embedding space



https://nlp.stanford.edu/projects/glove/

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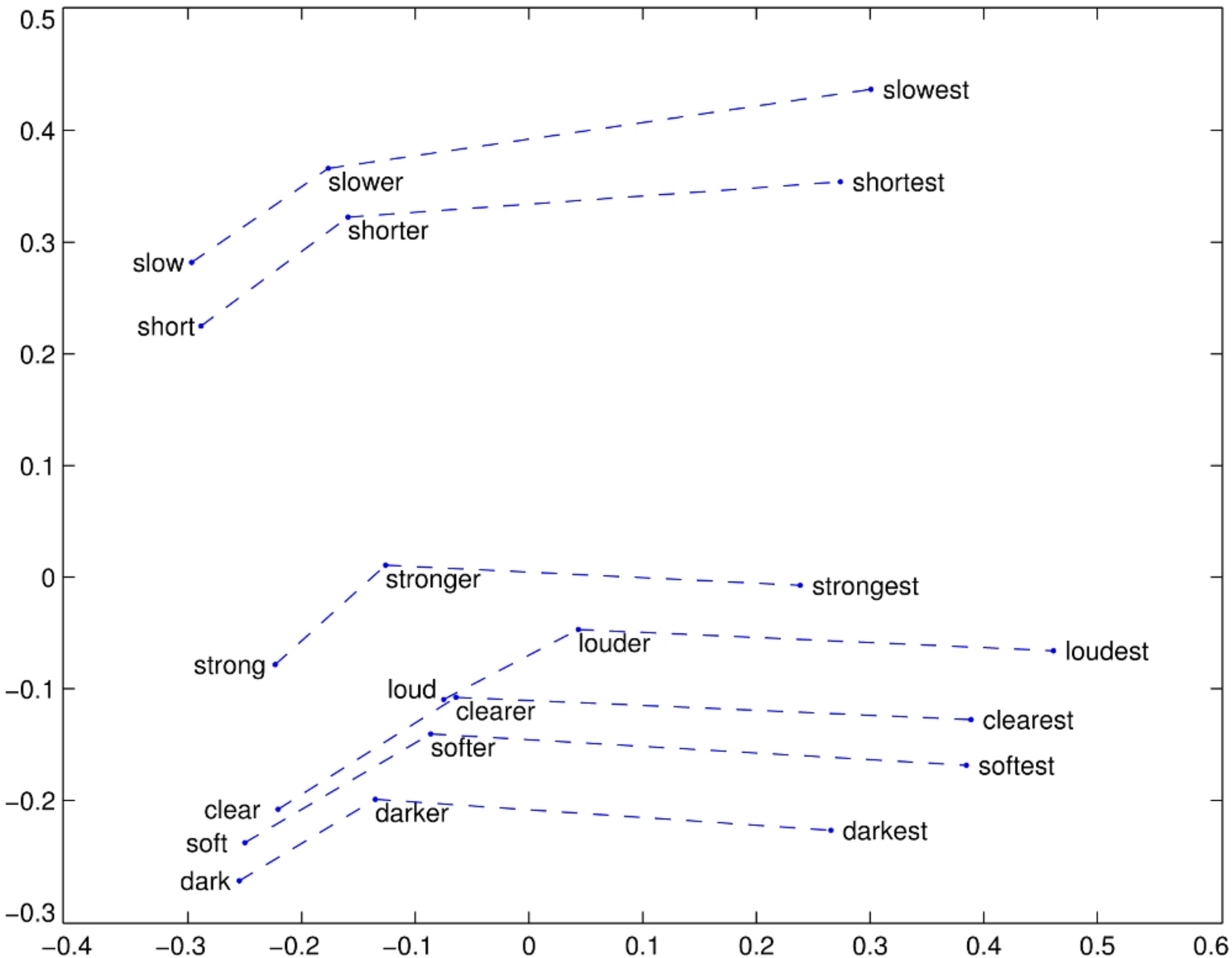


What can we do with embeddings?

- Unsupervised *transfer learning* from a large corpus

- Document embeddings
  - 1. Supervised learning: Bag-of-Embeddings logreg
    - labeled train docs->labeled new docs
  - 2. Unsupervised learning / exploratory analysis
    - docs->[analysis]

- Wordlist-based inferences
  - 3. Semi-automatic dictionary expansion
    - (words->words)

(1) Sup. learning with document embedding

- Instead of bag-of-words, can we derive a latent embedding of a document/sentence?
  - "Bag of embeddings" or "averaged word embeddings" representation
  - You can use it just like a BOW logistic regression - it's just a different type of feature vector
  - Pros/cons?
- Especially for shorter texts, BoE LR typically outperforms BOW LR.

See: Arora et al. 2017

(2) Unsup. learning with document embedding

- Example: tweets about mass shootings (Demszky et al. 2019)
  1. Average word embeddings => tweet embeddings
  2. Cluster tweets (k-means)
  3. Interpret clusters' words (closest to centroid)

Topic	10 Nearest Stems
news (19%)	break, custodi, #breakingnew, #updat, confirm, fatal, multipl, updat, unconfirm, sever
investigation (9%)	suspect, arrest, alleg, apprehend, custodi, charg, accus, prosecutor, #break, ap
shooter's identity & ideology (11%)	extremist, radic, racist, ideolog, label, rhetor, wing, blm, islamist, christian
victims & location (4%)	bar, thousand, california, calif, among, los, southern, veteran, angel, via
laws & policy (14%)	sensibl, regul, requir, access, abid, #gunreformnow, legisl, argument, allow, #guncontolnow
solidarity (13%)	affect, senseless, ach, heart, heartbroken, sadden, faculti, pray, #prayer, deepest
remembrance (6%)	honor, memori, tuesday, candlelight, flown, vigil, gather, observ, honour, capitol
other (23%)	dude, yeah, eat, huh, gonna, ain, shit, ass, damn, guess

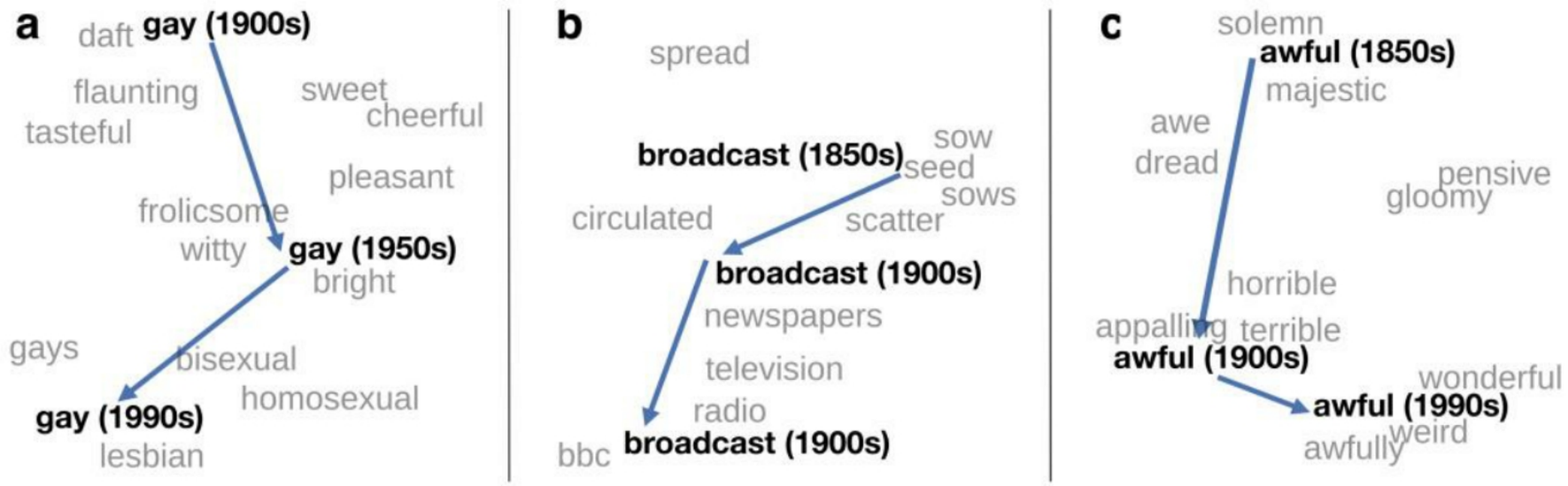
Table 1: Our eight topics (with their average proportions across events) and nearest-neighbor stem embeddings to the cluster centroids. Topic names were manually assigned based on inspecting the tweets.

Application: keyword expansion

- I have a few keywords for my task. Are there any I missed?
- Automated or semi-automated new terms from embedding neighbors

- Other non-embedding lexical resources can do this too (e.g. WordNet), but word embeddings typically cover a *lot* of diverse vocabulary

Word use over time
[Hamilton et al. 2016]



Embeddings over time

- A note on aligning spaces: This is nontrivial
  - Straightforward with count-based spaces, where dimensions are words
  - Prediction-based spaces, where dimensions cannot be interpreted: Dimension 5 in two different spaces doesn't need to be the same thing
  - But the spaces encode similar kinds of contexts, so we can align them
  - Method: "orthogonal Procrustes"
    - Best possible rotational alignment of dimensions

Embeddings over time

- Diachronic lexical semantics:
  - Hamilton, Lescovec and Jurafsky 2016, Diachronic Word Embeddings Reveal Statistical Laws of Semantic Change, https://aclanthology.org/P16-1141/
  - Del Tredici, Fernandez, Boleda 2019: Short-Term Meaning Shift: A Distributional Exploration, https://aclanthology.org/N19-1210/
  - Detecting changes in word senses.
  - Changes in context, while the sense stays the same: noise
- Diachronic social analysis:
  - Kutuzov, Velldal, Øvrelid 2017, Tracing armed conflicts with diachronic word embedding models https://aclanthology.org/W17-2705.pdf
  - Detecting changes in usage context
  - Changes in word sense: noise
- **One researcher's signal is another researcher's noise**